GATE-ALL BRANCHES

ENGINEERING MATHEMATICS

PROBABILITY AND STATISTICS

Lecture No.- 11



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Topics covered in previous lecture



Topic

Probability density function

Topic

Question based on RV

Topics to be Covered



Topic Expectation of random variable.

Topic Variance and Standard Deviation of random variable.

Topic Question based on expectation,variance and standard deviation of the random variable.



Topic : Probability and Statistics

$f(x)$

#Q. $P_x(X) = Me^{(-2|x|)} + Ne^{(-3|x|)}$ is the probability density function for the real random variable X, over the entire x-axis, M and N are both positive real numbers. The equation relating M and N is-

$f_X(x) = Me^{-2|x|} + Ne^{-3|x|}$ X = continuous random variable

M and N Are constants

A $M + 2/3\ N = 1$

B $2M + 1/3\ N = 1$

C $M + N = 1$

D $M + N = 3$

$$f_x(x) = Me^{-2|x|} + Ne^{-3|x|}$$

$$\int_{-\infty}^{\infty} f_x(x) = 1$$

$$\int_{-\infty}^{\infty} R_x(x) = 1$$

Valid prob. Density Function

$$\int_{-\infty}^{\infty} \left(Me^{-2|x|} + Ne^{-3|x|}\right) dx = 1$$

x = simple function

$|x|$ = modulas function

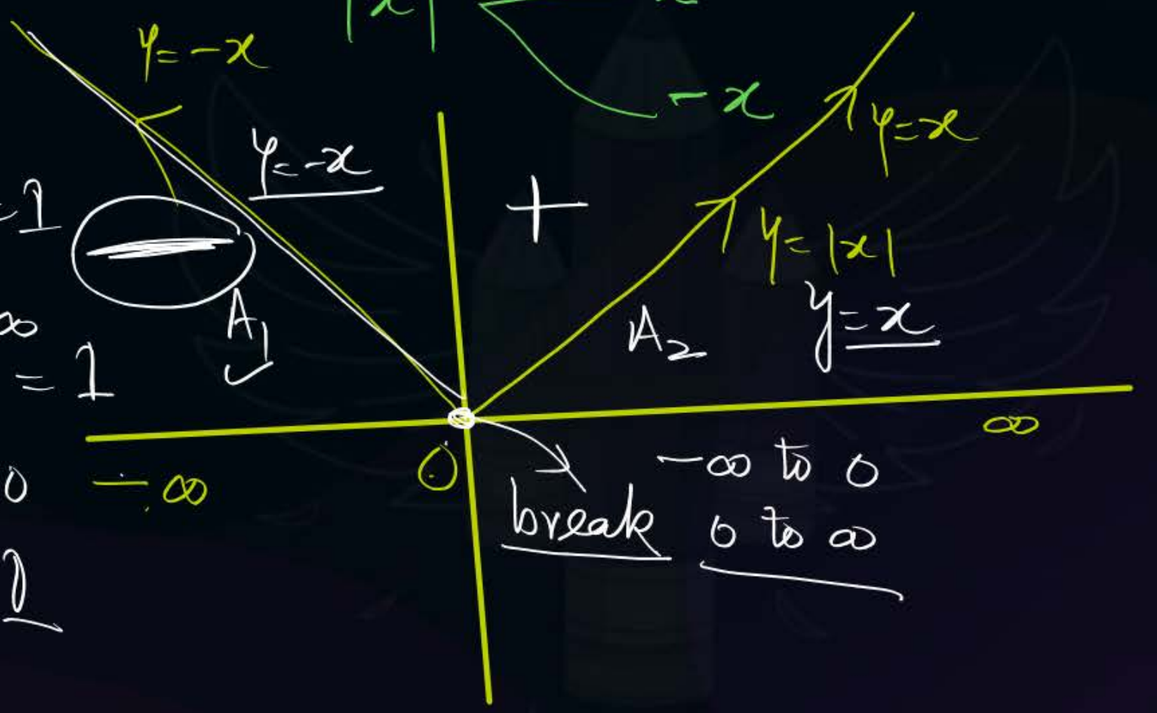
= compound function

$$= \int_{-\infty}^{0} \left[Me^{-2(-x)} + Ne^{-3(-x)}\right] dx + \int_{0}^{\infty} \left[Me^{-2(x)} + Ne^{-3(x)}\right] dx = 1$$

$$\int_{-\infty}^{0} Me^{2x} + Ne^{3x}\, dx + \int_{0}^{\infty} Me^{-2x} + Ne^{-3x}\, dx = 1$$

$$\left[\frac{Me^{2x}}{2} + \frac{Ne^{3x}}{3}\right]_{-\infty}^{0} + \left[\frac{Me^{-2x}}{-2} + \frac{Ne^{-3x}}{-3}\right]_{0}^{\infty} = 1$$

$$\left[\frac{M}{2} + \frac{N}{3} + 0\right] + \left[0 - \left(-\frac{M}{2} - \frac{N}{3}\right)\right] = 1$$



Expectation of a random variable:

(A) Average / weighted mean / Expectation / center of mass / center of gravity)

mean value of a random variable

Large No. of Trials

MEAN = center value → center value

(Data Points Are Balanced)

Average ⇒ Small No. of Trials

mean = Large No. of Trials

20000 child marks Dis

Children	Marks
Rahul(30)	20
Deepak(30)	19
	20
	21
	22
	23

Convert to The Single value.

Information

X	fi
x_1	f_1
x_2	f_2
x_3	f_3
x_4	f_4
x_n	f_n

→ prob. Relative Frequency

$$\text{mean} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

mean

$$= \frac{M}{2} + \frac{N}{3} + \frac{M}{2} + \frac{N}{3} = 1$$

$$= \boxed{M + \frac{2N}{3} = 1}$$

M and N are constants

Median : Mid value

If data is even ⟶ $\frac{n}{2}$ th, $\frac{n}{2}+1$ th term

If data is odd ⟶ $\frac{n+1}{2}$ th Term

Mode: Highest Frequent value

Center value
Center Measure
of Tendency

2 3 4 5 6 1 2 12 15 9 7 6 7

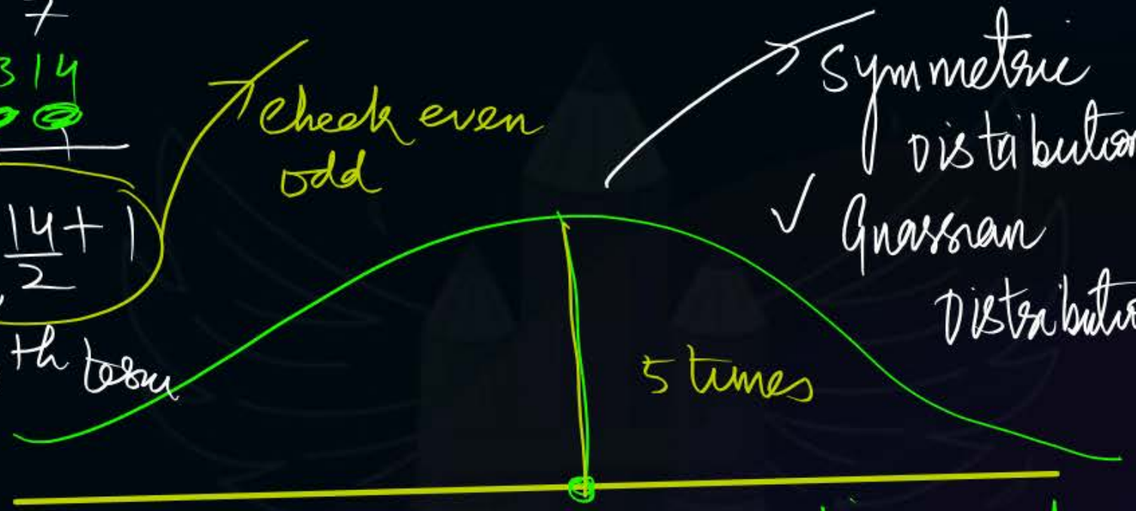
1 2 3 4 5 6 7 8 9 10 11 12 13 14



mid value = median = even = $\frac{14}{2}, \frac{14+1}{2}$

= 7, 8 th term

Check even odd



$$\text{mean} = \frac{\text{Weight balance}}{\text{mean}}$$

$$\frac{2+3+4+5+6+1+2+12+15+9+7+6+7}{14}$$

Discrete Random variable: (Countable Infinite SET)

X	x_0	x_1	x_2	x_3		x_n
P(X=x)	p_0	p_1	p_2	p_3		p_n

$E[X] = \mu$ = mean of Random variable X

$$E[X] = \frac{x_0 p_0 + x_1 p_1 + x_2 p_2 + \cdots + x_n p_n}{p_0 + p_1 + \cdots + p_n}$$

$$E[X] = x_0 p_0 + x_1 p_1 + x_2 p_2 + \cdots + x_n p_n$$

$$E[X] = \sum_{i=0}^{n} x_i p_i$$

expected value

Throwing A Die

X = No. of dots

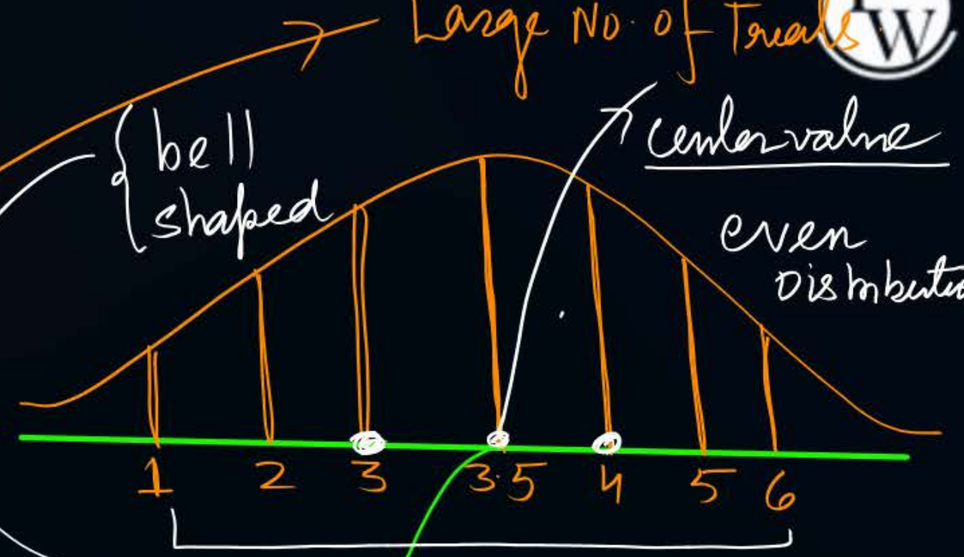
$X = 1, 2, 3, 4, 5, 6$

X	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E[X] = \mu = 1\times\frac{1}{6} + 2\times\frac{1}{6} + 3\times\frac{1}{6} + 4\times\frac{1}{6} + 5\times\frac{1}{6} + 6\times\frac{1}{6}$$

$$= \frac{1+2+3+4+5+6}{6}$$

$$\boxed{E[X] = 3.5}$$



mean = 3.5
median = 3.5
mode = 3.5
(Normal distribution)

$\frac{6}{2}, \frac{6}{2}+1$

3, 4

$\Rightarrow \frac{3}{=}$ or (4)

Playing A Game

X	1	2	3	4	5	6
Payoff	−3	+1	+1	+1	+2	−3
	Loose	win	win	win	win	Loose

① ① ① ①

$X = +1, +2, -3$ (random variable)

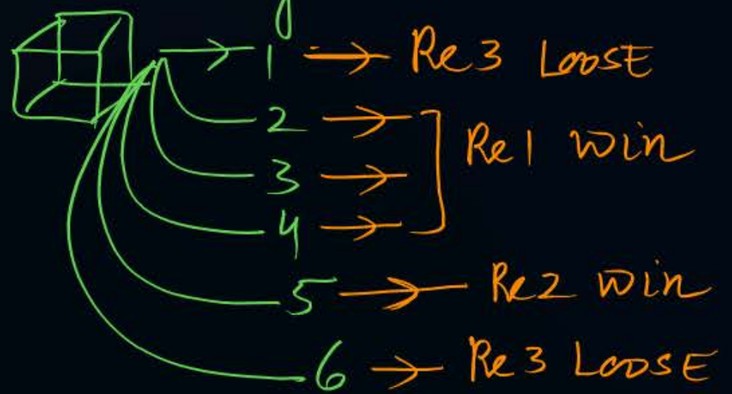
$P(\text{winning}) = \frac{4}{6} = \frac{2}{3}$

$P(\text{loosing}) = \frac{2}{6} = \frac{1}{3}$

$P(\text{wining}) \uparrow$

$P(\text{loosing}) \downarrow$

Throwing A Die



Should we Play This game?

For Large of Trials

X	+1	+2	−3
P(X=x)	3/6	1/6	2/6

For large No. of Trials

$$\text{Total Payoff} = 1 \times \frac{3}{6} \times n + 2 \times \frac{1}{6} \times n + (-3) \times \frac{2}{6} \times n$$

$n = 100$
$n = 1000$
$n = 100000$

$$\underline{1} \times \frac{3}{6} \times n + 2 \times \frac{1}{6} \times n + (-3) \times \frac{2}{6} \times n = -\frac{n}{6}$$

$$\frac{n}{2} + \frac{n}{3} - n = -\frac{n}{6}$$

$$x_1 \times P[X=x_1] \times n + x_2 P[X=x_2] \cdot n + x_3 P[X=x_3] \cdot n = -\frac{n}{6}$$

$$= \cancel{n}\left[x_1 P[X=x_1] + x_2 P[X=x_2] + x_3 P[X=x_3]\right] = \frac{-\cancel{n}}{6}$$

$$= x_1 P[X=x_1] + x_2 P[X=x_2] + x_3 P[X=x_3] = -\frac{1}{6}$$

$$= \sum_{i=1}^{3} x_i P[X=x_i] = -\frac{1}{6}$$

WM X
LOOSE ✓

Mean / expectation = −Neg

Quit this game

[Casino, always wm]

Expected value $= \sum_{i=0}^{n} x_i p_i$

Variance: $V(X) = E[X^2] - [E[X]]^2$

$\Sigma f_i x_i^2 - (\Sigma f_i x_i)^2$

For Discrete Random variable"

X	x_0	x_1	x_2		x_n
$P(X=x)$	p_0	p_1	p_2		p_n

$E[X^2] = x_0^2 p_0 + x_1^2 p_1 + x_2^2 p_2 + \dots + x_n^2 p_n$

$$E[X^2] = \sum_{i=0}^{n} x_i^2 p_i$$

$$V(X) = \sum_{i=0}^{n} x_i^2 p_i - \left[\sum x_i p_i\right]^2$$

Variance can't be Negative

Standard deviation $= \sqrt{\text{variance}} = \sqrt{E[X^2] - [E[X]]^2}$

$V(X) > 0$

Continuous Random variable:

✓ $\text{MEAN} = E[X] = \int_{-\infty}^{\infty} x f(x)\,dx \qquad -\infty \le x \le \infty$

If Interval is given $a \le x \le b$

$$E[X] = \int_a^b x f(x)\,dx$$

$$E[X^2] = \int_a^b x^2 f(x)\,dx$$

$$E[X^3] = \int_a^b x^3 f(x)\,dx$$

$$E[X^n] = \int_a^b x^n f(x)\,dx$$

✓ $\text{variance} = V(X) = E[X^2] - [E[X]]^2$

$$V(X) = \int_a^b x^2 f(x)\,dx - \left[\int_a^b x f(x)\right]^2$$

✓ standard deviation $= \sqrt{\text{variance}}$



Topic :Probability and statistics

0, 1, 2 defective
discrete

X	0	1	2
P(X=x)	1/6	2/3	1/6

#Q. A machine produces 0, 1, or 2 defective pieces in a day with associated probability of 1/6, 2/3 and 1/6 respectively. Then mean value and the variance of the number of defective pieces produced by

MEAN value $= \sum_{i=0}^{n} x_i p_i = 0 \times \frac{1}{6} + 1 \times \frac{2}{3} + 2 \times \frac{1}{6} = \frac{2}{3} + \frac{1}{3} = 1$

(A) 1 and 1/3

(B) 1/3 and 1

(C) 1 and 4/3

(D) 1/3 and 4/3

$= \frac{2}{3} + \frac{4}{6} - (1)^2$

$= \frac{1}{3}$

Variance $= E[X^2] - [E[X]]^2 = (0)^2 \frac{1}{6} + (1)^2 \times \frac{2}{3} + (2)^2 \frac{1}{6} - (1)^2$

$\frac{24}{18} - (1)^2$

$\frac{12}{9} - (1)^2$

$= \frac{1}{3}$ $\frac{4}{3} - (1)^2$



Topic :Probability and statistics

HH 2
HT 1
TH 1
TT 0

Letter Form → word

#Q. Each of the nine words in the sentence "The Quick brown fox jumps over the lazy dog" is written on a separate piece of paper. There nine pieces of paper are kept in a box. One of the pieces is drawn at random form the box. The expected length of the word drawn is____.
(The answser should be rounded to one decimal place.)

"The Quick brown fox jumps over The Lazy Dog"

	①	②	③	④	⑤	⑥	⑦	⑧	⑨
No. of letters	3	5	5	3	5	4	3	4	3

Frequency

$X = 3, 4, 5$

X	3	4	5
$P(X=x)$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{3}{9}$

$E[X] = 3\times\frac{4}{9} + 4\times\frac{2}{9} + 5\times\frac{3}{9}$

$= \frac{35}{9} = 3.88$



Topic :Probability and statistics

#Q. A random variable X has probability density function f(x) as given below:

$$f(x)=\begin{cases} a+bx & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

If the expected value E[X] = 2/3,then Pr[X < 0.5] is_____.



Topic :Probability and statistics

#Q. Consider the following probability mass function (p.m.f) of a random variable X.

$$p(x,q)=\begin{cases} q & \text{if } X=0 \\ 1-q & \text{if } X=1 \\ 0 & \text{otherwise} \end{cases}$$

Discrete Points

X	0	1
P(X=x)	q	1-q

if q = 0.4, the variance of X is _____.

Variance $= E[X^2]-[E[X]]^2$

$E[X^2]=(0)^2\times q+(1)^2\times(1-q)$

$=1-q=1-0.4=0.6$

$E[X]=0\times q+1\times 1-q=1-q=1-0.4$

$=0.6$

$V(X)=E[X^2]-[E[X]]^2$

$=0.6-(0.6)^2$

$=0.6-0.36=0.24$

THANK - YOU